

Ansh Agnihotri

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Education

IIT Senapati Manipur

Dec 2022 – Present

B.Tech in Electronics & Communication Engineering

CGPA: 7.53/10

- Relevant Coursework: **Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks**, Database Management Systems (DBMS)

Experience

Zidio Development

May 2025 – June 2025

Data Science Intern

Remote

- Engineered a scalable fraud detection service using an **XGBoost** model to process over **590K** financial transactions, achieving **95.9%** balanced accuracy.
- Designed and implemented a resilient ETL pipeline to automate data ingestion and preprocessing, significantly accelerating data preparation for model training.
- Developed and deployed a user-facing web application with Streamlit, enabling stakeholders to interactively monitor and interpret real-time model predictions.

Projects

HealthMate | Flask, LangChain, Groq API, Pinecone, Docker, AWS ECR, GitHub Actions [\[GitHub\]](#)

June 2025

- Developed a medical AI chatbot using the **Groq Llama 3.1 8B** model with a RAG architecture over a corpus of medical book **Gale of Encyclopedia**, achieving **95%** accuracy in information retrieval.
- Containerized the full-stack application using **Docker** and implemented a **CI/CD pipeline** with GitHub Actions to automate the build and deployment process.
- Integrated Pinecone to manage over **100,000** vector embeddings (384-dimensional), enabling efficient similarity searches with cosine metrics via a REST API.

AI-Powered Data Science Interviewer | Flask, FAISS, MongoDB, EdgeTTS [\[Demo\]](#)

July 2025

- Designed a real-time voice-based mock interviewer leveraging **Llama 3 on Groq**, achieving **sub-second (<0.8s) speech-to-speech latency** for natural conversational flow.
- Developed a **RAG pipeline with FAISS** to retrieve and synthesize insights from 100+ pages of ML literature, producing **technically accurate, non-repetitive interview questions**.
- Implemented a production-ready **Flask backend** with Groq Whisper transcription, EdgeTTS synthesis, MongoDB authentication, and LangSmith tracing, ensuring secure, scalable, and monitored deployment.

Audio Classification Using SNN | Python, PyTorch, snnTorch, Librosa [\[GitHub\]](#)

Aug 2025 – Sep 2025

- Developed a Spiking Neural Network for environmental sound classification, training on the **UrbanSound8K dataset** (**8,732** audio clips across 10 classes).
- Achieved **63.19% test accuracy**, with a Macro ROC-AUC of **0.817** and a Micro ROC-AUC of **0.837**, demonstrating the model's effectiveness.

Loan Default Prediction | Python, FastAPI, Gradient Boosting, PostgreSQL [\[Code\]](#)

Aug 2024 – Nov 2024

- Developed a production-ready machine learning system to predict loan repayment risk, integrating diverse behavioral and temporal data streams.
- Engineered robust feature pipelines from GPS logs, in-app events, and masked attributes, ensuring strict prevention of data leakage.
- Implemented Gradient Boosting models with stratified sampling, achieving **ROC-AUC 0.61+** and strong PR-AUC performance on imbalanced lending data.

Technical Skills

- Languages:** Python, C++, SQL, JavaScript, HTML/CSS
- Backend & DevOps:** Flask, FastAPI, Docker, Git, REST APIs, GitHub Actions, AWS (ECR, EC2)
- Databases:** MongoDB, Pinecone, FAISS
- Machine Learning & Data Science:** PyTorch, TensorFlow, Scikit-learn, LangChain, Pandas, NumPy
- Developer Platforms:** Streamlit, Render, Power BI

Certifications & Achievements

Tata Group GenAI Powered Data Analytics Job Simulation (Forage) [\[Certificate\]](#)

Top 5% (Gold) in NPTEL Fundamentals of AI (IIT Madras) among 4,000+ candidates [\[Certificate\]](#)

Elite (Top 10%) in NPTEL Deep Learning (IIT Ropar) [\[Certificate\]](#)

Top 12% Globally in Kaggle Backpack Price Prediction competition; outperformed 3,200+ participants.